

Legend - Hierarchical machine learning identifies genomic features underlying serotype variation in *Vibrio cholerae*

Figure:

Figure 1: Population structure analysis of 1,866 *Vibrio cholerae* genomes (A) Distribution of VSP-I genomic islands among the PCoA clusters **(B)** Distribution of VSP-II genomic islands among the PCoA clusters **(C)** Distribution of serotypes among the PCoA clusters **(D)** Core genome phylogenetic tree with adjacent bars indicating the serotypes, PCoA clusters, VSP-I and II genomic islands, respectively

Figure 2: Features for hierarchical machine learning-based classification. (A) and **(C)** show the distribution of ROC-AUC values obtained by the logistic regression and random forest classifiers using feature sets selected by Boruta, LASSO, and their intersection for serogroup and serotype classification, respectively. **(B)** and **(D)** show the SHAP feature importance plots of the best-performing model for serogroup and serotype classification, respectively. Genes located within the OAGC regions are highlighted in blue.

Figure 3: Heatmaps depicting the distribution of significantly associated (A) AMR, (B) virulence genes and (C) integrons among serotypes grouped based on PCoA clustering

Table:

Table 1: Model performance for serogroup classification before and after 10 fold CV and hyperparameter tuning

Table 2: Model performance for serotype classification before and after 10 fold CV and hyperparameter tuning

Supplementary material:

Figures:

Supplementary Figure 1 (Additional file-1): Mapping of misclassified genomes (false positives and false negatives) for (A) serogroup and (B) serotype classification.

Tables:

Supplementary table 1 (Additional file-1): VSP genomic island classification criteria

Supplementary table 2 (Additional file 2):

Sheet 1: Chisq test association between AMR genes and serotype

Sheet 2: Chisq test association between virulence genes and serotype

Sheet 3: Chisq test association between integrons and serotype

Sheet 4: Chisq test association between transposons and serotype